Weekly Putnam

11 October 2026

PROBLEM

Let $z_1, z_2, \ldots, z_{2027}$ be complex numbers of absolute value 1 such that

$$z_1^2 + z_2^2 + \cdots + z_{2027}^2 = 0.$$

Determine, with proof, the largest possible value of

$$|z_1 + z_2 + \cdots + z_{2027}|.$$

Answer: $1 + 45\sqrt{1013}$ · **Level:** A4–A5 · **Topic:** Complex numbers, extremal inequality

Hints

1. Rotate all the z_k so that their sum is real, and write $z_k = x_k + iy_k$. What does the hypothesis say about $\sum x_k^2$? The Cauchy–Schwarz inequality then gives $2027/\sqrt{2}$. Which configurations would give equality, and why can an odd number of points never form one?
2. The squares z_k^2 are 2027 unit vectors with sum zero. Sort them by the sign of their imaginary part. One side holds more vectors than the other, yet both sides have the same total imaginary part in absolute value. How much must the larger side give up?
3. The cheapest way to pay is to place a single z_k at ± 1. To prove that nothing cheaper exists, pair the remaining vectors across the two sides and apply Cauchy–Schwarz twice: once to the sums within the pairs, once to the differences.

The key idea

> With the sum rotated onto the positive real axis, the identity $x = \frac{1}{\sqrt{2}}\left(x^2 + \frac{1}{2}\right) - \frac{1}{\sqrt{2}}\left(x - \frac{1}{\sqrt{2}}\right)^2$ turns the problem into an exact one: the sum equals $2027/\sqrt{2}$ minus a multiple of $Q = \sum (x_k - 1/\sqrt{2})^2$, so one must show that the real parts cannot all sit close to $1/\sqrt{2}$. The obstruction is parity. The unit vectors z_k^2 have zero sum, and an odd number of them cannot split evenly between the upper and lower half-planes, so the larger side must lose a full unit of imaginary part. No single-variable inequality sees this, because the same point would have to be penalised on one side and rewarded on the other; the proof has to couple the two sides.

Solution

Throughout, $n = 2027 = 2m + 1$ with $m = 1013$, and $M = 1 + \sqrt{m(2m-1)} = 1 + \sqrt{1013 \cdot 2025} = 1 + 45\sqrt{1013}$.

The value M is attained. Let $a = \sqrt{(2m-1)/(4m)}$ and $b = \sqrt{(2m+1)/(4m)}$, so that $a^2 + b^2 = 1$. Take one of the numbers equal to 1, m of them equal to $a + ib$ and m equal to $a - ib$. The sum of the squares is $1 + 2m(a^2 - b^2) = 1 - 1 = 0$, and the sum is $1 + 2ma = 1 + \sqrt{m(2m-1)} = M$.

Step 1: normalisation. Multiplying every z_k by the same complex number of absolute value 1 changes neither the hypothesis nor $|z_1 + \cdots + z_n|$, so we may assume that $z_1 + \cdots + z_n$ is real and nonnegative. Write $z_k = x_k + iy_k$. Then $|z_1 + \cdots + z_n| = \sum x_k \le \sum |x_k|$. Replacing z_k by $-z_k$ does

not change z_k^2, so it suffices to prove

$$X := \sum_k x_k \le M \qquad \text{whenever } x_k \ge 0,\ x_k^2 + y_k^2 = 1,\ \sum_k z_k^2 = 0.$$

Replacing every z_k by its conjugate changes nothing either. Since n is odd we may therefore assume that the set $P = \{k : y_k \ge 0\}$ has more elements than $N = \{k : y_k < 0\}$. The real and imaginary parts of the hypothesis read

$$\sum_k (x_k^2 - y_k^2) = 0, \qquad \sum_k x_k y_k = 0, \qquad \text{hence} \qquad \sum_k x_k^2 = \frac{n}{2}. \tag{1}$$

Step 2: an exact formula. For every real x,

$$x = \frac{1}{\sqrt{2}}\Big(x^2 + \frac{1}{2}\Big) - \frac{1}{\sqrt{2}}\Big(x - \frac{1}{\sqrt{2}}\Big)^2.$$

Summing over k and using (1),

$$X = \frac{n}{\sqrt{2}} - \frac{Q}{\sqrt{2}}, \qquad Q := \sum_k \Big(x_k - \frac{1}{\sqrt{2}}\Big)^2. \tag{2}$$

Let Q^* be the value of Q for the configuration above. That configuration satisfies (1), so (2) gives $M = n/\sqrt{2} - Q^*/\sqrt{2}$. Hence the theorem is equivalent to

$$Q \ge Q^*. \tag{3}$$

Step 3: new variables. Put

$$\sigma_k = y_k^2 - x_k^2 = 1 - 2x_k^2 \in [-1, 1], \qquad \kappa_k = 2x_k|y_k| \in [0, 1], \qquad \sigma_k^2 + \kappa_k^2 = 1.$$

(These are, up to sign, the real and imaginary parts of z_k^2.) By (1),

$$\sum_k \sigma_k = 0, \qquad \sum_{k \in P} \kappa_k = \sum_{k \in N} \kappa_k. \tag{4}$$

Since $x_k = \sqrt{(1 - \sigma_k)/2}$, we have $Q = \sum_k g(\sigma_k)$, where

$$g(\sigma) = \tfrac{1}{2}\big(1 - \sqrt{1 - \sigma}\big)^2 = \frac{\sigma^2}{2\big(1 + \sqrt{1 - \sigma}\big)^2}, \qquad -1 \le \sigma \le 1.$$

In the extremal configuration $\sigma = -1$ once and $\sigma = 1 - 2a^2 = \frac{1}{2m}$ for the other $2m$ points, so

$$Q^* = c + G, \qquad c := g(-1) = \frac{3 - 2\sqrt{2}}{2} = 0.0857\ldots, \qquad G := 2m\, g\Big(\frac{1}{2m}\Big).$$

We shall use the following facts about g.

(g1) $g(\sigma) \ge c\,\sigma^2$ for all $\sigma \in [-1, 1]$, and $g(\sigma) \ge \sigma^2/8$ for $\sigma \ge 0$. Both follow from the second expression for g, since $1 + \sqrt{1 - \sigma} \le 1 + \sqrt{2}$ always and $1 + \sqrt{1 - \sigma} \le 2$ for $\sigma \ge 0$.

(g2) g is convex: $g'(\sigma) = \frac{1}{2}\big((1 - \sigma)^{-1/2} - 1\big)$ and $g''(\sigma) = \frac{1}{4}(1 - \sigma)^{-3/2} > 0$ on $[-1, 1)$. In particular $g'(-1) = -\frac{2 - \sqrt{2}}{4} = -0.146\ldots$

(g3) $G \le \dfrac{1}{12m}$. Indeed, for $0 \le \varepsilon \le \frac{1}{4}$ we have $1 - \sqrt{1 - \varepsilon} = \dfrac{\varepsilon}{1 + \sqrt{1 - \varepsilon}} \le \dfrac{\varepsilon}{2 - \varepsilon}$, so $g(\varepsilon) \le \dfrac{\varepsilon^2}{2(2 - \varepsilon)^2} \le \dfrac{\varepsilon^2}{6}$; now take $\varepsilon = \frac{1}{2m}$.

Step 4: parity. Let $\delta_k = 1 - \kappa_k \in [0,1]$. Then

$$\sigma_k^2 = 1 - \kappa_k^2 = \delta_k(2 - \delta_k) \geq \delta_k. \tag{5}$$

By (4),

$$\sum_{k\in P} \delta_k = |P| - \sum_{k\in P} \kappa_k = |P| - \sum_{k\in N} \kappa_k = |P| - |N| + \sum_{k\in N} \delta_k \;\geq\; |P| - |N| \;\geq\; 1. \tag{6}$$

This is the only place where the parity of n enters. Fix an index in P for which κ is smallest and call it 0; thus $\delta_0 \geq \delta_k$ for all $k \in P$. Write $P' = P \setminus \{0\}$.

Step 5: the easy cases. By (g1), $Q \geq c \sum_{k\in P} \sigma_k^2$.

Case (i): $\kappa_0 \geq \frac{1}{2}$. Then $\delta_k \leq \frac{1}{2}$ for every $k \in P$, so $\sigma_k^2 \geq \frac{3}{2}\delta_k$ by (5), and by (6) $Q \geq \frac{3}{2}c > c + G$, because $c/2 > 1/(12m) \geq G$.

Case (ii): $\kappa_0 < \frac{1}{2}$ and $c\,\kappa_0(1-\kappa_0) \geq G$. By (5) and (6), $\sum_{k\in P'} \sigma_k^2 \geq \sum_{k\in P'} \delta_k \geq 1 - \delta_0 = \kappa_0$, while $\sigma_0^2 = 1 - \kappa_0^2$. Hence $Q \geq c\,(1 - \kappa_0^2 + \kappa_0) \geq c + G$.

In the remaining cases $\kappa_0 < \frac{1}{2}$ and $c\,\kappa_0(1-\kappa_0) < G$. Then $\frac{1}{2}\kappa_0 < \kappa_0(1-\kappa_0) < G/c \leq 1/(12cm)$, so

$$\kappa_0 < \frac{1}{6cm} < \frac{2}{m}. \tag{7}$$

Case (iii): (7) holds and $\sigma_0 \geq 0$. Then $Q \geq g(\sigma_0) \geq \sigma_0^2/8 = (1 - \kappa_0^2)/8 > \frac{1}{9} > c + G$.

Step 6: the main case. Now (7) holds and $\sigma_0 = -s$ with $s = \sqrt{1 - \kappa_0^2} \in (0,1]$. Note that $1 - s \leq \kappa_0^2$. Suppose, for contradiction, that $Q < Q^*$.

(a) All other points are nearly balanced. By convexity, $g(-s) \geq g(-1) + g'(-1)(1-s) \geq c - \frac{1}{6}\kappa_0^2 > c - \frac{2}{3m^2}$. Therefore

$$\sum_{k\neq 0} g(\sigma_k) < Q^* - g(\sigma_0) < G + \frac{2}{3m^2} \leq \frac{1}{12m} + \frac{2}{3m^2} < \frac{1}{11m},$$

and by (g1)

$$\sum_{k\neq 0} \sigma_k^2 < \frac{1}{11cm} < \frac{1.06}{m} < \frac{1}{900}.$$

So $|\sigma_k| < \frac{1}{30}$ and $\kappa_k = \sqrt{1 - \sigma_k^2} > 0.95$ for every $k \neq 0$.

(b) The two sides are as balanced as parity allows. By (5), $\sum_{k\in P'} \delta_k \leq \sum_{k\neq 0} \sigma_k^2 < 1$, while (6) gives $\sum_{k\in P'} \delta_k \geq |P| - |N| - 1$. As $|P| - |N|$ is odd and positive, $|P| - |N| = 1$. Thus $|P'| = |N| = m$. Number the elements of P' and of N from 1 to m in any way, and write σ_i, κ_i for the ith element of P' and σ_i', κ_i' for the ith element of N.

(c) Two balance relations. Put $u_i = \sigma_i + \sigma_i'$, $w_i = \sigma_i - \sigma_i'$, $V = \sum_i u_i^2$ and $W = \sum_i w_i^2$. The first relation in (4) says $\sum_i u_i = s$, so by Cauchy–Schwarz

$$V \geq \frac{s^2}{m}. \tag{8}$$

The second relation in (4) says $\kappa_0 + \sum_i \kappa_i = \sum_i \kappa_i'$, that is,

$$\kappa_0 = \sum_i (\kappa_i' - \kappa_i) = \sum_i \frac{\kappa_i'^2 - \kappa_i^2}{\kappa_i + \kappa_i'} = \sum_i \frac{\sigma_i^2 - \sigma_i'^2}{\kappa_i + \kappa_i'} = \sum_i \frac{u_i w_i}{\kappa_i + \kappa_i'}.$$

Since $\kappa_i + \kappa_i' > 1.9$, Cauchy–Schwarz gives $\kappa_0 \leq \sqrt{VW}/1.9$, hence

$$V\,W \geq 3.6\,\kappa_0^2. \tag{9}$$

(d) Strong convexity. On the interval $J = [-\frac{1}{30}, \frac{1}{30}]$ we have $g'' \geq \frac{1}{4}(31/30)^{-3/2} > \frac{1}{5}$. Let $\varepsilon = s/(2m)$, which lies in J. By Taylor's formula, for $\sigma \in J$,

$$g(\sigma) \geq g(\varepsilon) + g'(\varepsilon)(\sigma - \varepsilon) + \tfrac{1}{10}(\sigma - \varepsilon)^2.$$

Sum this over the $2m$ indices $k \neq 0$. The linear terms cancel, because $\sum_{k\neq 0} \sigma_k = s = 2m\varepsilon$. Moreover

$$\sum_{k\neq 0} (\sigma_k - \varepsilon)^2 = \frac{1}{2}\sum_i \big((u_i - 2\varepsilon)^2 + w_i^2\big) = \frac{1}{2}\Big(V - \frac{s^2}{m}\Big) + \frac{1}{2}W.$$

Hence, with $\Psi(t) := g(-t) + 2m\, g\big(\frac{t}{2m}\big)$ for $0 \leq t \leq 1$,

$$Q \;\geq\; \Psi(s) + \frac{1}{20}\Big(V - \frac{s^2}{m} + W\Big). \tag{10}$$

(e) Conclusion. We have $\Psi(1) = Q^*$, and Ψ is convex because g is. Its derivative at 1 is

$$\Psi'(1) = -g'(-1) + g'\Big(\frac{1}{2m}\Big) = \frac{2-\sqrt{2}}{4} + \frac{1}{2}\Big(\big(1 - \tfrac{1}{2m}\big)^{-1/2} - 1\Big) < \frac{1}{5}.$$

So $\Psi(s) \geq \Psi(1) - \Psi'(1)(1-s) \geq Q^* - \frac{1}{5}\kappa_0^2$, and (10) becomes

$$Q \;\geq\; Q^* + \frac{1}{20}\Big(V - \frac{s^2}{m} + W - 4\kappa_0^2\Big).$$

It remains to see that the bracket is nonnegative. Both $V - s^2/m$ and W are nonnegative, by (8). If $V - s^2/m \geq 4\kappa_0^2$ we are done. Otherwise, by (7), $V < \frac{1}{m} + \frac{16}{m^2} < \frac{2}{m}$, and then (9) gives $W \geq 3.6\,\kappa_0^2/V \geq 1.8\,m\,\kappa_0^2 \geq 4\kappa_0^2$. In either case $Q \geq Q^*$, contradicting the assumption.

Thus (3) holds in every case, and by (2) $X \leq n/\sqrt{2} - Q^*/\sqrt{2} = M$. Together with the example at the start, the largest possible value is

$$1 + \sqrt{1013 \cdot 2025} = 1 + 45\sqrt{1013} \approx 1433.2447.$$

Why it is hard, and why it is original

Why it is hard. The bound $2027/\sqrt{2} \approx 1433.3055$ is immediate, and the true maximum $1 + 45\sqrt{1013} \approx 1433.2447$ lies only 0.06 below it. Equality in Cauchy–Schwarz needs every z_k^2 to be $\pm i$, which is possible for an even number of points and impossible for an odd number. Every argument that treats the points one at a time (tangent lines, Jensen, Lagrange multipliers with a convex certificate) returns $2027/\sqrt{2}$ again, because the constraint that matters is a parity constraint. The extremal configuration is not a small perturbation of the balanced one: a single point jumps to $z = 1$, where it is a local minimum of the natural Lagrangian, and the other 2026 points move by an amount of order $1/2026$.

The year. For an odd number n of points the same configuration gives $1 + \sqrt{(n-1)(n-2)/2}$, and numerical optimisation returns this value for every odd n from 3 to 13. For $n = 2027$ the radicand is $1013 \cdot 2025 = 45^2 \cdot 1013$. The proof above uses the size of 2027 only to keep the numerical constants simple.

Relatives. The nearest classical relative is Problem 1 of the 1973 International Mathematical Olympiad: an odd number of unit vectors in a half-plane has a sum of length at least 1. The parity step here is the same phenomenon, but that problem asks for nothing quantitative beyond it. There is also a companion question with the roles exchanged: if $z_1 + \cdots + z_{2027} = 0$, then the largest value of $|z_1^2 + \cdots + z_{2027}^2|$ is $2025 - \frac{1}{1013}$, by the same method with the quadratic function σ^2 in place of g. The present problem is the harder of the two, since g is not symmetric and the margin below the trivial bound is far smaller.

Verification

- The extremal configuration for $n = 2027$ was evaluated with 50-digit arithmetic: the sum of squares vanishes to within 10^{-47}, and the modulus of the sum agrees with $1 + 45\sqrt{1013} = 1433.24474165555936943\ldots$ to all 50 digits.
- Constrained numerical maximisation of $|z_1 + \cdots + z_n|$ subject to $z_1^2 + \cdots + z_n^2 = 0$ (sequential quadratic programming, 500 random starting points for each n) gave $1 + \sqrt{(n-1)(n-2)/2}$ to eight decimal places for $n = 3, 5, 7, 9, 11, 13$, and $n/\sqrt{2}$ for $n = 4, 6$.
- For $n = 2027$, 1803 feasible configurations were generated by perturbing the extremal one at scales from 10^{-4} to 0.2 (including moving the special point freely) and restoring the constraint exactly. None exceeded the claimed maximum; the closest fell short by 6.7×10^{-6}.
- Every numerical constant used in the proof was checked directly for $m = 1013$: $c = 0.0857864$, $G = 6.1713 \times 10^{-5} < 1/(12m)$, $1/(6c) = 1.9428$, $1/(11c) = 1.0597$, $\min_J g'' = 0.2380$, $\Psi'(1) = 0.14657$.
- The identity $\sum x_k = n/\sqrt{2} - Q/\sqrt{2}$ was confirmed on the extremal configuration: both sides equal $1433.2447416555\ldots$